

$$x^T \begin{bmatrix} \end{bmatrix} \Bigg\}^m$$

$$A > 0 \Leftrightarrow x^T A x > 0, \forall x \neq 0$$

$$\begin{bmatrix} \end{bmatrix} \begin{bmatrix} \end{bmatrix} \begin{bmatrix} \end{bmatrix} = \begin{bmatrix} \end{bmatrix}$$

$$A > 0 \Leftrightarrow \lambda_i(A) > 0$$

$$A = U \Delta U^T = \sum_{i=1}^n \lambda_i \mu_i \mu_i^T$$

$$\begin{bmatrix} \mu_i \end{bmatrix} \begin{bmatrix} \lambda_i \end{bmatrix} \begin{bmatrix} \mu_i^T \end{bmatrix} = \begin{bmatrix} \mu_i \mu_i^T \end{bmatrix}$$

$$x^T A x > 0, \forall x \neq 0$$

$$x^T \left(\sum_{i=1}^n \lambda_i \mu_i \mu_i^T \right) x = \sum_{i=1}^n \lambda_i \underbrace{x^T \mu_i}_c \underbrace{\mu_i^T x}_c = \sum_{i=1}^n \lambda_i \underbrace{(x^T \mu_i)}_{>0}^2_{>0} > 0$$

$$c = x^T \mu_i \in \mathbb{R} \quad c = c^T = (x^T \mu_i)^T = \mu_i^T x$$

$$\begin{bmatrix} \mu_i^T \end{bmatrix} \begin{bmatrix} \mu_i \end{bmatrix} = \begin{bmatrix} \end{bmatrix}$$

$$A = U \Delta U^T; \quad \Delta = \text{diag}(\lambda_1, \dots, \lambda_n)$$

$$A^{-1} = U \Delta^{-1} U^T; \quad \Delta^{-1} = \text{diag}(\lambda_1^{-1}, \dots, \lambda_n^{-1})$$

$$A A^{-1} = U \underbrace{\Delta U^T U}_{I} \underbrace{\Delta^{-1} U^T}_{A^{-1}} = U \underbrace{\Delta \Delta^{-1}}_I U^T = U U^T = I$$

$$A^{-1} A = I$$

$$A \succ 0 \Rightarrow \text{tr}(A) > 0$$

$$\sum_{i=1}^n \lambda_i > 0$$

$\uparrow$
 $\lambda_i > 0$

$$A \succ 0 \Rightarrow \det(A) > 0$$

$$\prod_{i=1}^n \lambda_i > 0$$

$\uparrow$
 $\lambda_i > 0$

Problem 4

$$A = BB^T \succeq 0$$

$$x^T A x \geq 0$$

$$x^T A x = x^T (BB^T) x = \underbrace{x^T B}_{y^T} \underbrace{B^T x}_y = \begin{pmatrix} y^T B^T x \\ y^T x^T B \end{pmatrix} = y^T y = \|y\|_2^2 \geq 0$$

Problem 5

$U = A(A^T A)^{-1/2}$ is an orthogonal matrix

$$U^T U = I$$

$$U^T = [A(A^T A)^{-1/2}]^T = [(A^T A)^{-1/2}]^T A^T = (A^T A)^{-1/2} A^T$$

$$\underbrace{(A^T A)^{-1/2}}_{C^{-1/2}} \underbrace{A^T A}_C \underbrace{(A^T A)^{-1/2}}_{C^{-1/2}} = I$$

$$\begin{aligned} C &= C^H, \quad B = C^{1/2}, \quad B = B^T \\ BB^T &= C^{1/2} C^{1/2} = C \\ B^{-1} B B^T B^{-T} &= B^{-1} C B^{-T} \\ &= I = C^{-1/2} C C^{-1/2} \end{aligned}$$